

An Analysis of Bayesian network

Bayesian network

$$P(h \mid D) = \frac{P(D \mid h)P(h)}{P(D)}$$

Bayesian network -- Part 1

Developing Bayesian networks Developing a Bayesian network often begins with creating a DAG G such that X satisfies the local Markov property with respect to G . Sometimes this is a causal DAG. The conditional probability distributions of each variable given its parents in G are assessed. In many cases, in particular in the case where the variables are discrete, if the joint distribution of X is the product of these conditional distributions, then X is a Bayesian network with respect to G .

Bayesian network -- Part 2

$\Pr(R = T \mid G = T) = 0.00198 \frac{T T T + 0.1584 T F T + 0.00198 T T T + 0.288 T T F + 0.1584 T F T + 0.0 T F F}{891} = \frac{891}{2491} \approx 35.77\%$

Bayesian network -- Part 3

with the term $\Pr(S = T \mid R)$ removed, showing that the action affects the grass but not the rain. These predictions may not be feasible given unobserved variables, as in most policy evaluation problems. The effect of the action $\text{do}(x)$ can still be predicted, however, whenever the back-door criterion is satisfied. It states that, if a set Z of nodes can be observed that d-separates (or blocks) all back-door paths from X to Y then

Bayesian network -- Part 4

A back-door path is one that ends with an arrow into X . Sets that satisfy the back-door criterion are called "sufficient" or "admissible." For example, the set $Z = R$ is admissible for predicting the effect of $S = T$ on G , because R d-separates the (only) back-door path $S \leftarrow R \rightarrow G$. However, if S is not observed, no other set d-separates this path and the effect of turning the sprinkler on ($S = T$) on the grass (G) cannot be predicted from passive observations. In that case $P(G \mid \text{do}(S = T))$ is not "identified". This reflects the fact that, lacking interventional data, the observed dependence between S and G is due to a causal connection or is spurious (apparent dependence arising from a common cause, R). (see Simpson's paradox) To determine whether a causal relation is identified from an arbitrary Bayesian network with unobserved variables, one can use the

three rules of "do-calculus" and test whether all do terms can be removed from the expression of that relation, thus confirming that the desired quantity is estimable from frequency data. Using a Bayesian network can save considerable amounts of memory over exhaustive probability tables, if the dependencies in the joint distribution are sparse. For example, a naive way of storing the conditional probabilities of 10 two-valued variables as a table requires storage space for $2^{10} = 1024$ values. If no variable's local distribution depends on more than three parent variables, the Bayesian network representation stores at most $10 \cdot 2^3 = 80$ values. One advantage of Bayesian networks is that it is intuitively easier for a human to understand (a sparse set of) direct dependencies and local distributions than complete joint distributions.

Bayesian network -- Part 5

Marginal independence structure In general, learning a Bayesian network from data is known to be NP-hard. This is due in part to the combinatorial explosion of enumerating DAGs as the number of variables increases. Nevertheless, insights about an underlying Bayesian network can be learned from data in polynomial time by focusing on its marginal independence structure: while the conditional independence statements of a distribution modeled by a Bayesian network are encoded by a DAG (according to the factorization and Markov properties above), its marginal independence statements—the conditional independence statements in which the conditioning set is empty—are encoded by a simple undirected graph with special properties such as equal intersection and independence numbers.

Bayesian network -- Part 6

Given data x and parameter θ , a simple Bayesian analysis starts with a prior probability (prior) $p(\theta)$ and likelihood $p(x \mid \theta)$ to compute a posterior probability $p(\theta \mid x) \propto p(x \mid \theta) p(\theta)$. Often the prior on θ depends in turn on other parameters ϕ that are not mentioned in the likelihood. So, the prior $p(\theta)$ must be replaced by a likelihood $p(\theta \mid \phi)$, and a prior $p(\phi)$ on the newly introduced parameters ϕ is required, resulting in a posterior probability

Bayesian network -- Part 7

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This is the simplest example of a hierarchical Bayes model. The process may be repeated; for example, the parameters ϕ may depend in turn on additional parameters ψ , which require their own prior. Eventually the process must terminate, with priors that do not depend on unmentioned parameters.

Bayesian network -- Part 8

Using the expansion for the joint probability function $\Pr(G, S, R)$ and the conditional probabilities from the conditional probability tables (CPTs) stated in the diagram, one can evaluate each term in the sums in the numerator and denominator. For example,

Bayesian network -- Part 9

To answer an interventional question, such as "What is the probability that it would rain, given that we wet the grass?" the answer is governed by the post-intervention joint distribution function

Bayesian network -- Part 10

$\Pr(R, G \mid \text{do}(S = T)) = \Pr(R) \Pr(G \mid R, S = T)$

Bayesian network -- Part 11

Causal networks Although Bayesian networks are often used to represent causal relationships, this need not be the case: a directed edge from u to v does not require that X_v be causally dependent on X_u . This is demonstrated by the fact that Bayesian networks on the graphs:

Bayesian network -- Part 12

Further reading Conrady S, Jouffe L (2015-07-01). Bayesian Networks and BayesiaLab – A practical introduction for researchers. Franklin, Tennessee: Bayesian USA. ISBN 978-0-9965333-0-0. Charniak E (Winter 1991). "Bayesian networks without tears" (PDF). AI Magazine. Kruse R, Borgelt C, Klawonn F, Moewes C, Steinbrecher M, Held P (2013). Computational Intelligence A Methodological Introduction. London: Springer-Verlag. ISBN 978-1-4471-5012-1. Borgelt C, Steinbrecher M, Kruse R (2009). Graphical Models – Representations for Learning, Reasoning and Data Mining (Second ed.). Chichester: Wiley. ISBN 978-0-470-74956-2.